

HTR Research Lab — Complete User Guide

Location: [/research-lab](#) **Access:** Professional plan and above **Purpose:** Six interactive quantitative modeling labs — 19 tools total — covering payment models, policy simulation, population health, AI governance, FHIR interoperability, and knowledge management. All computations run entirely in the browser. No patient data or inputs are transmitted to any server.

Overview of the Six Labs

Lab	URL	Tools
Payment Models & VBC	/research-lab/payment-models	APM Design Lab, APM Shared Savings Calculator, CEA Calculator
Policy & Quality Sciences	/research-lab/policy-quality	Policy Simulator, Clinical Quality Optimizer, Hospital Financial Scorecard, HTA Studio, Actuarial Lab
Population & Equity	/research-lab/population-equity	Population Health Modeler, AI Analytics Lab (Digital Health)
Technology & AI	/research-lab/technology-ai	AI Performance Analyzer, Algorithmic Bias Auditor, AI Governance Builder, Build vs. Buy Calculator, AI Scribe ROI
Interoperability & Risk	/research-lab/interoperability	FHIR Interoperability Lab, Risk Stratification Engine
Knowledge & Workspace	/research-lab/knowledge-workspace	Evidence Library, Workforce Modeler, Innovation Leaderboard, Research Workspace

Part I — Payment Models & VBC

[/research-lab/payment-models](#)

Three tools for designing, stress-testing, and evaluating alternative payment models from global budgets to episode bundles to cost-effectiveness analysis.

Tool 1 — APM Design Lab

A three-tab modeling studio for designing alternative payment models at multiple levels of complexity.

Tab 1 — ACO Architecture Designer

Design a custom ACO financial model by configuring all structural parameters and immediately seeing the resulting financial waterfall and viability assessment.

Section A — Model Structure

Input	Options	Notes
Model Type	FFS with Shared Savings, Episode-Based, Prospective Capitation, Global Budget, Hybrid	Drives which financial mechanics apply
Risk Arrangement	One-Sided (upside only), Two-Sided (upside + downside), Full Risk	One-sided has no loss exposure; two-sided unlocks higher sharing rates

Input	Options	Notes
Attribution Method	Claims-Based, Panel-Based, Hybrid	How patients are assigned to the ACO
Benchmark Methodology	Regional (CMS Regional), National, Blended (Regional + National)	Blended shows a regional/national weight slider

Section B — Financial Parameters

Input	Range	Description
Upside Sharing Rate	0–100%	ACO's share of gross savings above MSR
Downside Sharing Rate	0–100%	ACO's share of losses below MLR (two-sided only)
Minimum Savings Rate (MSR)	0–5%	Savings threshold ACO must exceed before sharing begins
Minimum Loss Rate (MLR)	0–5%	Loss threshold before loss sharing is triggered
Savings Cap	0–20%	Maximum shared savings as % of benchmark
Loss Cap	0–20%	Maximum loss payment as % of benchmark
Quality Withhold	0–10%	% of shared savings held pending quality performance
Quality Threshold Score	0–100	Minimum quality score to recover withhold

Section C — Population Data

Input	Range
Attributed Lives	500–50,000
Benchmark PMPM (\$)	\$400–\$2,500
Actual Spend (% of benchmark)	75%–120%
Quality Performance Score	0–100

Section D — Results Dashboard

Output	Description
Gross Savings vs Benchmark	Total dollar savings; % of benchmark
Net ACO Position	After MSR, sharing rate, withhold, and caps
PMPM Equivalent	Net position divided by attributed lives and months
Break-Even Spend	The actual spend % at which the ACO breaks even
Financial Waterfall	Bar chart: Benchmark → Actual Spend → MSR Gate → Sharing Applied → Quality Withhold → Net Position
Viability Badge	Green (viable), Amber (marginal), Red (not viable) with reason text

Model Flags — Three flags show pass/fail in real time:

- **MSR Gate** — whether savings exceed the minimum savings rate threshold
- **Downside Exposure** — maximum dollar loss if two-sided and under benchmark
- **Quality Withhold** — whether quality score meets threshold to recover withheld funds

Tab 2 — Episode-Based Payment Designer

Design bundled payment models for specific procedural episodes.

Episode Types (8 pre-configured)

Episode	30-Day Base Cost	90-Day Base Cost	180-Day Base Cost
Joint Replacement	\$22,000	\$28,000	\$32,000
CABG	\$45,000	\$55,000	\$62,000
Hip Fracture	\$24,000	\$32,000	\$38,000
Pneumonia	\$14,000	\$18,000	\$22,000
Major Bowel Procedure	\$35,000	\$42,000	\$50,000
Cesarean Section	\$16,000	\$20,000	\$24,000
PCI	\$28,000	\$34,000	\$40,000
Chemotherapy	\$18,000	\$30,000	\$52,000

Configuration Inputs

- **Episode Duration** — 30, 60, 90, or 180 days
- **Services Included** — checkboxes for 8 service categories (Facility Fee 38%, Physician Fee 18%, Anesthesia 7%, Post-Acute SNF 15%, Home Health 6%, Rehab Therapy 5%, Readmissions 7%, Labs/Imaging 4%); each shows its % of total cost
- **Target Price Method** — CMS-Based (historical −3%), Market-Based (−5%), Negotiated (custom % discount slider)
- **Episode Volume** — 10–2,000 episodes/year
- **Gainsharing Distribution** — Hospital %, Physician %; Reinvestment % auto-calculates to 100%

Outputs

- Current Average Episode Cost and Target Bundle Price
- Per-Episode Variance (positive = under target = savings)
- Total Gainsharing Pool (variance × volume)
- Gainsharing bar chart showing Hospital / Physician / Reinvestment splits
- Loss sharing alert if episodes exceed target
- Episode Cost Waterfall by care setting showing each service component

Tab 3 — Global Budget Simulator

Model a 5-year spending trajectory under a global budget cap vs. unconstrained healthcare trend.

Inputs

Input	Range	Default
Global Budget Growth Rate (annual)	0–6%	3.5%
Base Year Total Spending	\$500M–\$10,000M	\$2,000M
Population Growth	−1% to +3%	0.5%
General Healthcare Inflation	3–8%	5.0%
Hospital Cost Trend	2–8%	4.5%
Physician Cost Trend	2–7%	3.5%
Pharmaceutical Trend	5–15%	8.0%
SDOH Investment Offset toggle	On/Off	Off
Trend Reduction from SDOH Investment	0.5–2.0%	1.0%

Composite Unconstrained Trend is calculated as a weighted average:

- Hospital: 45% weight
- Physician: 30% weight
- Pharmaceutical: 15% weight
- Other/Inflation: 10% weight

5-Year Projection Output — for each year:

Output
Capped Budget (compound growth at your rate)
Unconstrained Spend (composite trend x population growth)
Required Efficiency (gap as % of unconstrained)
Feasibility status: Green (< 3%), Amber (3–5%), Red (> 5% — Feasibility Alert)

Tool 2 — APM Shared Savings Calculator

A simpler, preset-driven calculator for five standard ACO models. Ideal for rapid what-if analysis without configuring every structural parameter.

Model Presets

Model	MSR	Sharing Rate	Loss Share	Quality Withhold	Savings Cap
MSSP Track 1	2.0%	50%	None	0%	10%
MSSP Enhanced	0.0%	75%	40%	5%	15%
ACO REACH	0.0%	50%	50%	5%	20%
BPCI-Advanced	0.0%	100%	100%	0%	None
Custom	All adjustable				

Inputs (all models)

Input	Range
Attributed Lives	500–50,000
Benchmark PMPM (\$)	\$400–\$2,500
Actual Spend (% of benchmark)	75%–120%
Quality Performance Score	0–100
Admin Cost PMPM (\$)	\$0–\$50

Outputs

Output	Description
Annual Benchmark	Total benchmark spending
Gross Savings	Benchmark minus actual spend
Net Savings after MSR	Gross savings less minimum savings threshold
Quality Multiplier	Multiplier based on quality score (< 40 = 0.0x, 40–59 = 0.5x, 60–79 = 0.75x, ≥ 80 = 1.0x)
Shared Savings (capped)	Net savings × sharing rate × quality multiplier, capped at savings cap
Loss Payment	Downside payment if applicable
Admin Cost	Admin PMPM × attributed lives × 12

Output	Description
Net Position	Shared savings – admin cost – loss payment
Net Position PMPM	Net position divided by attributed lives and 12 months
Break-Even Savings %	The gross savings % at which net position = 0

Tool 3 — Cost-Effectiveness Analysis (CEA) Calculator

Calculate ICER (Incremental Cost-Effectiveness Ratio) and classify an intervention against four standard willingness-to-pay (WTP) thresholds.

Pre-built Condition Presets (12 conditions) — clicking a condition pre-fills all inputs with evidence-based defaults:

Condition	Example Pre-Set Cost/Patient	Efficacy Rate
Cardiovascular Disease	(varies by intervention)	varies
Diabetes T2	(prevention program defaults)	varies
Mental Health	(collaborative care defaults)	varies
Cancer Screening	(colonoscopy defaults)	varies
Opioid Use Disorder	(MOUD defaults)	varies
COPD		
Hypertension		
Stroke Prevention		
Obesity		
HIV Prevention		
Asthma		
Maternal Health		

Inputs

Input	Description
Cost per Patient (\$)	Annual cost of the new intervention per patient
Efficacy Rate (%)	% of treated patients who respond
QALY Gain per Responder	Quality-adjusted life years gained
Population Size	Number of eligible patients
Time Horizon (years)	1–20 years
Discount Rate (%)	0–10%; applied to future QALYs
Comparator Cost per Patient (\$)	Annual cost of standard of care

Outputs

Output	Formula
Responders	Population × Efficacy Rate
Number Needed to Treat (NNT)	100 / Efficacy Rate
Total Intervention Cost	Cost per Patient × Population

Output	Formula
Incremental Cost	(Intervention – Comparator) × Population
Total QALYs Gained	Responders × QALY Gain
Discounted QALYs	Total QALYs adjusted for discount rate and time horizon
ICER	Incremental Cost ÷ Discounted QALYs

ICER Classification

ICER Range	Classification
< \$0 (dominant)	Highly Cost-Effective — Dominant
\$0 – \$30,000	Highly Cost-Effective (NICE threshold)
\$30,001 – \$100,000	Cost-Effective (ICER Standard threshold)
\$100,001 – \$150,000	Borderline (ICER High threshold)
\$150,001 – \$200,000	High Value but Uncertain (CMS Informal threshold)
> \$200,000	Not Cost-Effective by standard thresholds

Part II — Policy & Quality Sciences

/research-lab/policy-quality

Five tools for policy simulation, quality measurement, hospital financial analysis, health technology assessment, and actuarial modeling.

Tool 1 — Policy Simulator

A four-tab simulator modeling Medicaid waiver design, all-payer global budgets, Medicaid expansion, and price transparency.

Tab 1 — 1115 Waiver Modeler

Design a Section 1115 Medicaid waiver and see projected enrollment, financial, population health, and CMS approval impacts.

Waiver Type (7 options with CMS Alignment Scores)

Waiver Type	CMS Alignment	Description
Global Commitment	High	Vermont-style global budget waiver
DSRIP (Delivery System Reform)	High	Hospital transformation waiver
Community Engagement	Moderate	Work/community activity requirements
Expansion Premium	Moderate	Premium assistance for marketplace coverage
Managed Care	High	MCO-based Medicaid delivery
Behavioral Health	High	BH services integration
Global Budget	Moderate	All-payer global budget variant

State Data (6 states pre-loaded)

State	Enrollees	Per-Capita Spending	FMAP	Uninsured Rate
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State	Enrollees	Per-Capita Spending	FMAP	Uninsured Rate
Vermont	218,000	\$9,800	56.87%	4.2%
New York	7,800,000	\$11,200	50.00%	6.8%
California	14,600,000	\$8,400	50.00%	7.2%
Texas	4,200,000	\$6,900	61.39%	18.4%
Ohio	3,100,000	\$8,100	63.68%	6.5%
Michigan	2,800,000	\$8,700	65.27%	5.8%

Waiver Design Parameters

Input	Range	Description
Coverage Expansion	0–25% of uninsured	% of currently uninsured newly covered
Managed Care Penetration	0–100%	% of enrollees in managed care
SDOH Investment Set-Aside	0–5%	% of total spending directed to social determinants
Premiums (above 138% FPL)	\$0–\$100/month	Monthly premiums for expansion population
Community Engagement Requirement	None / 20 hrs/week / 80 hrs/month	Work/community activity requirements
Benefit Limitations	None / Dental carve-out / Vision / Non-emergency transport	Service exclusions
Work Requirement	Toggle	Reduces approval likelihood by 25 points

Outputs

Output	Description
CMS Approval Likelihood	% probability (0–98%); color-coded bar; degraded by work requirements, high premiums
Enrollment Impact (5-Year)	New enrollees (expansion) / Lost enrollees (requirements) / Net change / Total Yr-5 enrollees
5-Year Spending Change	Total spend change; Federal share (90% FMAP for expansion); State net cost
Savings & Offsets	SDOH investment return (30¢ on dollar); Managed care efficiency (5% at 100% MC); Uncompensated care reduction (\$2,800/newly insured)
Hospital Financial Impact	Revenue gain from expansion; revenue loss from disenrollment; net hospital impact
Population Health Metrics	ER visit reduction; preventable hospitalization reduction; estimated lives saved (~1 per 455 newly insured)

Tab 2 — All-Payer Model / Global Budget Designer

10-year global budget projection for any payer configuration.

Inputs

Input	Range	Description
Base Year Spending	\$500M–\$20,000M	Total spending in base year
Annual Growth Cap	0–6%	Maximum allowed growth per year
Projected Unconstrained Trend	4–9%	What spending would grow to without a cap
Spending Allocation	Hospital 30–50% / Physician 28–35% / Drugs 10–20%	% of total budget by sector
Quality Pool	0–5%	% of budget reserved for quality payments
SDOH Pool	0–3%	% reserved for social determinants investment
Payer Alignment	Medicare / Medicaid / Commercial / Self-Pay toggles	Which payers are included in the global budget
VT ACO Compare Mode	Toggle	Overlays Vermont ACO model benchmarks

10-Year Projection Outputs — per year:

- Capped Budget vs. Trend Budget
- Annual Savings Generated
- Cumulative Savings
- Quality Allocation (\$)
- SDOH Allocation (\$)
- Required Productivity Improvement % (trend gap / growth rate)

Vermont ACO Benchmarks (shown when Compare Mode is on):

- Growth Target: 3.5% / Actual Growth: 4.2% / Quality Score: 78 / SDOH Investment: 1.2%

Tab 3 — Medicaid Expansion Impact

Non-expansion state analysis showing coverage gap and the impact of expansion.

Non-Expansion State Data (10 states pre-loaded): Texas, Florida, Georgia, Tennessee, Alabama, Mississippi, South Carolina, Kansas, Wisconsin, Wyoming — each with coverage gap size and annual uncompensated care costs.

Outputs:

- Coverage gap count and uninsured rate for selected state
- 5-year enrollment, spending, federal/state cost projections under expansion
- Uncompensated care reduction estimate

Tab 4 — Price Transparency Analyzer

Models the financial and utilization impact of CMS price transparency requirements.

Services with Rate Comparisons (10 procedures pre-loaded):

Service	HOPD Rate	ASC Rate	Office Rate
Knee Replacement	\$28,500	\$14,200	—
Colonoscopy	\$1,850	\$620	\$520
Echocardiogram	\$890	\$420	\$310

Service	HOPD Rate	ASC Rate	Office Rate
Infusion Therapy (per hour)	\$520	\$200	\$160
PT/OT (per session)	\$230	\$110	\$90
Lab Services (per panel)	\$180	\$95	\$60
(+ 4 additional procedures)			

NSA Surprise Billing Specialties: Emergency Medicine, Anesthesiology, Radiology, Pathology, Neonatology

Outputs: Potential savings per site-of-care migration; annual savings if X% of volume shifts to lower-cost setting; surprise billing exposure estimate.

Tool 2 — Clinical Quality Optimizer

Simulate HEDIS quality measure performance, CMS Star Ratings, MIPS composite scores, and Pay-for-Performance bonuses — with evidence-based improvement strategies for each measure.

Tab 1 — HEDIS Simulator

Model quality measure performance for a health plan or provider group.

Configuration: Organization Type (Health Plan / Provider Group); Plan Type (Commercial / Medicaid / Medicare Advantage)

15 HEDIS Measures with National Benchmarks

Measure	Code	P50 Benchmark	P90 Benchmark	Weight
Controlling High Blood Pressure	CBP	62%	74%	1.5
HbA1c Control (<8%)	HBD	60%	72%	1.5
Colorectal Cancer Screening	COL	55%	70%	1.3
Breast Cancer Screening	BCS	67%	79%	1.2
Childhood Immunization Status	CIS	66%	78%	1.2
Prenatal Care	PNC	83%	92%	1.2
Medication Adherence – Diabetes	MAH	80%	87%	1.2
BMI Assessment and Follow-Up	WCC	66%	80%	1.0
Cervical Cancer Screening	CCS	58%	71%	1.2
Depression Screening	DSF	67%	82%	1.1
Antidepressant Medication Mgmt	AMM	58%	70%	1.1
Follow-up after MH Hospitalization (7d)	FUH	39%	55%	1.4
Well-Child Visits	W34	70%	83%	1.0
Immunizations – Adolescents	IMA	37%	52%	1.0
Access to Preventive/Ambulatory	AAP	88%	95%	0.8

For each measure you can:

- Set your current performance with a slider
- See your 1–5 star rating vs. P50/P90 benchmarks
- Expand to see 4 evidence-based improvement strategies
- See opportunity score (gap to P90 × weight)

Composite HEDIS Score — weighted average of all measures; shown as a gradient card with overall star rating.

Top 5 Opportunities — sorted by weighted gap to P90; your highest-ROI measures ranked.

Tab 2 — CMS Star Ratings Simulator		
Model Medicare Advantage composite Star Rating across 5 domains (32 sub-measures).		
5 Star Rating Domains		
Domain	Weight	Measures
Staying Healthy	1.0x	8 measures including flu vaccine, cancer screenings, BMI, diabetes monitoring
Managing Chronic Conditions	3.0x	9 measures including BP control, diabetes, medication adherence (each 3x)
Member Experience (CAHPS)	4.0x	6 measures including getting needed care, appointments, customer service (each 4x)
Member Complaints & Appeals	1.5x	6 measures including timely decisions, appeals review, call center performance
Health Plan Administration	1.5x	3 measures including physical/mental health improvement, readmissions
Each measure has 5-star cut points. Enter your plan's performance on any measure and see the star contribution. Overall composite star rating drives CMS quality bonus payments.		
Tab 3 — MIPS Composite Score		
Model Merit-based Incentive Payment System (MIPS) composite score for physician practices.		
Four MIPS Categories		
Category	Weight	Input Method
Quality	30%	Select 6 quality measures; enter performance rates
Improvement Activities	15%	Select from 20 pre-built activities (10 or 20 points each)
Promoting Interoperability	25%	Score on 4 PI requirements
Cost	30%	Estimated based on resource use patterns
20 Improvement Activities pre-loaded (with point values):		
- Annual Psychosocial Needs Assessment (10 pts) - Care Coordination Agreements with Specialists (20 pts) - CMS-Approved Diabetes Prevention Program (20 pts) - PCMH Certification/Attestation (20 pts) - Behavioral Health Integration (20 pts) - FHIR & Interoperability Standards (10 pts) - Practice Improvements for Underserved Populations (20 pts) (+ 13 additional)		
Outputs: Composite MIPS score (0–100); payment adjustment (+/– % of Medicare payments); Exceptional Performance bonus threshold (≥ 75 points).		

Tab 4 — Pay-for-Performance ROI Calculator

Calculate annual P4P bonus revenue for a provider group.

12 P4P Measures with Base Bonus Values

Measure	Base Bonus
HbA1c Control (<8%)	\$850
Controlling High Blood Pressure	\$900
Follow-up after MH Hospitalization	\$1,100
Plan All-Cause Readmissions Reduction	\$1,400
Colorectal Cancer Screening	\$600
Breast Cancer Screening	\$600
Prenatal Care	\$750
Medication Adherence – Diabetes	\$700
Antidepressant Medication Management	\$650
Depression Screening	\$450
Well-Child Visits	\$400
Adolescent Immunizations	\$500

For each measure: enter your current performance rate and whether you participate. The calculator outputs total annual P4P bonus revenue and a per-provider average.

Tool 3 — Hospital Financial Scorecard

Benchmark your hospital's financial health against peer group medians. Stress-test against Medicaid cuts, volume changes, and travel nurse cost increases.

Peer Groups (4 options)

Peer Group	Median Operating Margin	Median Days Cash	Median DSCR	Median Current Ratio	Median Labor %
Critical Access Hospital (CAH)	0.8%	45 days	2.0×	1.8×	52%
Rural PPS Hospital	1.2%	58 days	2.5×	2.1×	50%
Urban Community Hospital	2.0%	90 days	3.2×	2.4×	48%
Urban Tertiary / Academic	3.1%	142 days	4.0×	2.8×	45%

Base Financial Inputs (7 sliders)

Input	Default	Range
Total Net Revenue	\$48M	\$5M–\$500M
Total Operating Expense	\$47.4M	\$5M–\$500M
Cash & Investments	\$5.8M	\$0–\$100M
Annual Debt Service	\$1.2M	\$0–\$50M
Current Assets	\$8.2M	\$1M–\$200M
Current Liabilities	\$4.1M	\$1M–\$200M
Total Labor Costs	\$24M	\$1M–\$300M

Stress Test Scenarios (3 sliders)

Scenario	Range	Mechanism
Medicaid Rate Cut	0–20%	Reduces revenue by: rate cut % × 15% (Medicaid share of net revenue)
Volume Change	–30% to +20%	Adjusts revenue proportionally
Travel Nurse Labor Increase	0–100%	Increases labor by: increase % × 12% (travel nurses' share of labor)

5 Scored Metrics

Metric	Scoring	Direction
Operating Margin	vs. peer median	Higher is better
Days Cash on Hand	vs. peer median	Higher is better
Debt Service Coverage Ratio	vs. peer median	Higher is better
Current Ratio	vs. peer median	Higher is better
Labor Cost % of Revenue	vs. peer median	Lower is better

Scoring Logic: Each metric is scored Strong ($\geq 115\%$ of benchmark), Adequate (85–115%), Weak (60–85%), or Critical ($< 60\%$).

Overall Status: Critical if ≥ 2 critical metrics; Weak if ≥ 1 critical or ≥ 2 weak; otherwise Adequate.

Benchmarks sourced from AHA Annual Survey 2024 and Kaufman Hall National Hospital Flash Report.

Tool 4 — HTA Studio (Health Technology Assessment)

Four-tab HTA toolkit: Budget Impact Model, Multi-Criteria Decision Analysis, Probabilistic Sensitivity Analysis, and Threshold Analysis.

Tab 1 — Budget Impact Model (BIM)

Model the 5-year payer budget impact of adopting a new intervention, including offsetting savings from avoided hospitalizations and ED visits.

Intervention Parameters

Input	Description
Intervention Name	Free text label
Target Condition	8 options: Diabetes T2, CHF, COPD, Depression, CKD, Oncology, Rare Disease, Hypertension
Total Population Size	Number of patients in the payer's covered population
Eligible Population (% of total)	% meeting clinical criteria for the intervention
Standard of Care Cost (annual/patient)	Annual cost of current therapy
New Intervention Cost (annual/patient)	Annual cost of new intervention
Market Uptake Scenario	Slow / Moderate / Rapid (see rates below)

Input	Description
Displacement (% switching from SoC)	% of eligible patients who switch
Total Covered Lives (for PMPM)	Used to calculate per-member-per-month cost impact

Uptake Scenario Rates

Scenario	Yr1	Yr2	Yr3	Yr4	Yr5
Slow	5%	15%	25%	32%	38%
Moderate	10%	25%	40%	50%	58%
Rapid	20%	40%	60%	72%	80%

Offsetting Savings (optional)

Offset Type	Inputs
Reduced Hospitalizations	Toggle; hospitalization reduction %; cost per hospitalization; baseline hospitalization rate
Reduced ED Visits	Toggle; ED reduction %; cost per ED visit; baseline ED visit rate

Outputs — per year and 5-year cumulative:

- Treated patients, drug cost, offsetting savings, net cost, cumulative cost
- 5-year total budget impact (base, low/slow, high/rapid scenarios)
- PMPM impact at base, low, and high scenarios
- Warning flag if total 5-year impact exceeds \$500M

Tab 2 — Multi-Criteria Decision Analysis (MCDA)

Evaluate interventions across clinical, economic, and equity criteria using weighted scoring.

Criteria Domains: Clinical Efficacy, Safety Profile, Cost-Effectiveness, Implementation Feasibility, Health Equity Impact, Quality of Evidence

Each domain is weighted and scored for each intervention alternative. Outputs a weighted composite score and radar chart for each alternative.

Tab 3 — Probabilistic Sensitivity Analysis (PSA)

Monte Carlo simulation (1,000 iterations) to model uncertainty in ICER estimates.

Inputs: Point estimates and standard deviations for cost and effectiveness parameters; distribution types (normal, log-normal, beta, gamma).

Outputs:

- Cost-effectiveness acceptability curve at multiple WTP thresholds
- Scatter plot of 1,000 cost-effectiveness pairs on the CE plane
- Probability cost-effective at each WTP threshold
- 95% credible interval for ICER

Tab 4 — Threshold & Surrogate Endpoint Analysis

Calculate the minimum clinical threshold required to achieve cost-effectiveness at a given WTP, and map surrogate endpoints (HbA1c, LDL reduction) to QALY estimates.

Inputs: Current cost, comparator cost, WTP threshold, current QALY gain.

Output: The minimum QALY gain or efficacy rate required for cost-effectiveness at the specified WTP.

Tool 5 — Actuarial Lab

Four-tab actuarial modeling suite for ACA plan design, premium rate setting, adverse selection dynamics, and IRA drug pricing.

Tab 1 — Actuarial Value (AV) Calculator

Estimate the actuarial value and metallic tier of an ACA-compliant health plan based on cost-sharing parameters.

Inputs (13 cost-sharing parameters)

Parameter	Default
Deductible	\$1,500
Out-of-Pocket Maximum	\$7,000
Coinsurance (%)	20%
PCP Copay	\$25
Specialist Copay	\$50
ER Copay	\$350
Urgent Care Copay	\$75
Generic Drug Copay	\$10
Preferred Brand Copay	\$45
Non-Preferred Brand Copay	\$95
Specialty Drug Coinsurance (%)	30%
Drug Deductible	\$0
CSR Silver Plan Toggle	Off

Algorithm: AV estimated from OOP max (50–80% base), adjusted for deductibles (–4% per \$1,000), coinsurance excess, and copay deltas from reference values. CSR toggle adds +7.3% AV (capped at 94%).

Outputs:

- Estimated AV % (45–97% range)
- Metallic Tier: Sub-Bronze (<56%), Bronze (56–68%), Silver (68–78%), Gold (78–88%), Platinum (≥88%)
- Tier visualization bar
- PMPM impact vs. standard Silver benchmark (~\$4 PMPM per 1% AV difference)

Tab 2 — Premium Rate Setting

Model the actuarially sound premium for a health plan.

Inputs: State, Plan Type (Individual/Small Group/Large Group), Target MLR, Expected Claims PMPM, Admin Load %, Profit Margin %, Risk Corridor toggle, Tobacco Surcharge %, Geographic Factor, Employer Contribution %.

Outputs: Required Premium PMPM; Employee share after employer contribution; Effective employee premium vs. benchmark comparison; MLR waterfall (claims → admin → profit → required premium).

Tab 3 — Adverse Selection Dynamics

Model how premium increases and policy design features drive enrollment exit and risk pool deterioration.

Inputs: Population size, Initial insured rate, Average health score, Community rating toggle, Risk adjustment method, Year-1 premium increase %, Exit rate per 10% premium increase, Market stabilizers (mandate, risk corridors, reinsurance, CSR payments).

Outputs:

- Enrollment trajectory over 5 years
- Risk pool health score over time
- Premium spiral (if adverse selection is occurring)
- Breakeven premium increase rate before death spiral risk

Tab 4 — IRA Drug Pricing Impact

Model the impact of the Inflation Reduction Act's Medicare drug price negotiation on 10 high-spend drugs.

Pre-loaded IRA Drugs (10)

Drug	Current List Price	MFP (post-negotiation)	Medicare Spending
Eliquis (apixaban)	\$7,120	44% of list	\$16.4B
Jardiance (empagliflozin)	\$6,450	34% of list	\$7.1B
Xarelto (rivaroxaban)	\$6,080	38% of list	\$6.1B
Januvia (sitagliptin)	\$6,200	21% of list	\$4.2B
Farxiga (dapagliflozin)	\$6,300	32% of list	\$3.1B
Entresto	\$8,200	39% of list	\$2.9B
Enbrel (etanercept)	\$74,000	25% of list	\$2.7B
Imbruvica (ibrutinib)	\$181,000	42% of list	\$2.4B
Stelara (ustekinumab)	\$119,000	34% of list	\$2.2B
Insulin aspart (NovoLog)	\$35	80% of list	\$1.8B

Inputs: Panel size (number of Medicare patients on each drug); toggle negotiated pricing on/off.

Outputs: Estimated savings per drug, total portfolio savings, PMPM impact on Medicare drug spend.

Part III — Population & Equity

/research-lab/population-equity

Tool 1 — Population Health Modeler

Four-tab simulation suite for chronic disease progression, epidemic modeling, population risk segmentation, and health equity analysis.

Tab 1 — Chronic Disease Progression Model

Markov chain simulation across a patient cohort over a 10-year horizon.

Disease Models

Disease	Severity States	Cost Range (annual/person)
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Disease	Severity States	Cost Range (annual/person)
Type 2 Diabetes	Pre-diabetes → Controlled → Uncontrolled → Complications → End-stage	\$2,400–\$48,000
Heart Failure	NYHA I → NYHA II → NYHA III → NYHA IV → End- stage/Transplant	\$8,500–\$120,000
COPD	GOLD 1 → GOLD 2 → GOLD 3 → GOLD 4 → Respiratory Failure	\$3,200–\$85,000
Chronic Kidney Disease	CKD 1-2 → Stage 3 → Stage 4 → Stage 5 → ESRD/Dialysis	\$3,800–\$92,000
Depression	Mild → Moderate → Mod-Severe → Severe → Treatment- Resistant	\$3,100–\$52,000

Inputs: Disease selection; cohort size (1,000–100,000); baseline state distribution (% across 5 states, must sum to 100%); Apply intervention toggle; Intervention efficacy (5–80% reduction in progression transitions); Cost per person per year (\$).

Pre-built Intervention Scenarios (15 total — 3 per disease)

Disease	Scenario	Efficacy	Cost/Person/Year
Diabetes	Intensive DM Program	42%	\$1,800
Diabetes	Community Health Worker	28%	\$900
Diabetes	CGM + Telehealth	35%	\$2,200
Heart Failure	Remote Patient Monitoring	38%	\$1,600
Heart Failure	ACE/ARB Optimization	25%	\$400
Heart Failure	Hospital-at-Home	45%	\$3,200
COPD	Pulmonary Rehab	32%	\$2,800
COPD	Care Transitions Program	22%	\$1,100
COPD	LABD Medication Optimization	28%	\$1,400
CKD	Nephrology Care Management	35%	\$2,400
CKD	SGLT2 Inhibitor Program	42%	\$3,600
CKD	Dietitian Integration	20%	\$800
Depression	Collaborative Care Model	38%	\$1,200
Depression	Digital CBT Platform	25%	\$600
Depression	PCMH BH Integration	32%	\$1,500

Outputs: 10-year stacked bar chart (cohort in each state over time); 10-year cumulative cost comparison (baseline vs. intervention); QALYs gained; ROI (net savings minus program cost); cost-per-QALY.

Tab 2 — SIR Epidemic Model

Susceptible-Infected-Recovered epidemiological model with a vaccination arm.

Inputs: Population size; Initial infected (%); Transmission rate (β); Recovery rate (γ); Vaccination coverage (%); Vaccine efficacy (%); Simulation duration (weeks).

Outputs: S/I/R trajectory chart over time; Peak infected count and week; Herd immunity threshold ($= 1 - 1/R_0$ where $R_0 = \beta/\gamma$); Whether vaccination coverage exceeds herd immunity threshold.

Tab 3 — Health Equity Studio

Model health disparity interventions across racial and socioeconomic dimensions.

Inputs: Baseline disparity gap (% outcome difference between highest and lowest groups); Intervention type; Population reach; Investment level.

Outputs: Projected disparity gap at Year 1, 3, 5; Lives affected; Cost per disparity-point-reduced.

Tab 4 — SDOH Impact Modeler

Quantify the health and financial return on SDOH investments (housing, food security, transportation, employment support).

Inputs: SDOH category; Population in need; Investment per person; Evidence-based impact factor.

Outputs: Health events avoided per year; 5-year medical cost offset; ROI multiple on SDOH investment.

Tool 2 — AI Analytics Lab (Digital Health)

Four-tab suite covering clinical AI performance, algorithmic bias, AI governance, build vs. buy, AI medical scribe ROI, and digital health technology ROI. (Full documentation in Part IV — Technology & AI.)

Part IV — Technology & AI

/research-lab/technology-ai

Tool 1 — AI Performance Analyzer

Four-tab clinical AI evaluation suite.

Tab 1 — Model Performance Evaluator

Enter raw confusion matrix data and calculate all standard clinical AI performance metrics.

Inputs: True Positives, False Positives, True Negatives, False Negatives (enter as counts or use sliders).

Outputs

Metric	Formula
Sensitivity (Recall)	$TP / (TP + FN)$
Specificity	$TN / (TN + FP)$
Precision (PPV)	$TP / (TP + FP)$
NPV	$TN / (TN + FN)$
F1 Score	$2 \times (Precision \times Recall) / (Precision + Recall)$
Accuracy	$(TP + TN) / Total$
MCC (Matthews Correlation Coefficient)	± 1 scale; robust for imbalanced datasets
AUROC	Estimated from sensitivity/specificity
Prevalence	$(TP + FN) / Total$

Alert fatigue index: $FP\ Rate \times (TP + FP) = \text{alerts per day if deployed}$. Color-coded: green (< 20 alerts/day), amber (20–50), red (> 50).

Tab 2 — Net Monetary Value (NMV) Calculator

Calculate the economic value of deploying a clinical AI model.

Inputs: True/False Positive costs, True/False Negative costs, deployment cost, patient volume, prevalence.

NMV Formula:

$$NMV = (TP \times TP_value) + (TN \times TN_value) - (FP \times FP_cost) - (FN \times FN_cost) - deployment_cost$$

Outputs: NMV per patient, annual NMV, break-even prevalence, optimal decision threshold recommendation.

Tab 3 — Clinical Threshold Optimizer

Find the optimal classification threshold for a given clinical use case by trading sensitivity against specificity.

Inputs: ROC curve data (or sensitivity/specificity pairs at multiple thresholds), cost ratio of FP:FN, target use case (screening vs. rule-out vs. treatment selection).

Outputs: Optimal threshold, expected sensitivity and specificity at that threshold, net clinical benefit curve.

Tab 4 — Deployment Readiness Checklist

Score a clinical AI model across 8 readiness dimensions: validation data, population representativeness, clinical workflow integration, monitoring plan, clinician training, consent and privacy, performance monitoring, and governance documentation.

Output: Readiness score (0–100); pass/fail by dimension; deployment risk rating (Low / Moderate / High).

Tool 2 — Algorithmic Bias Auditor

Three-tab tool for detecting and quantifying bias in clinical AI models.

Tab 1 — Fairness Criteria Tester

Test your model's performance data against three standard fairness criteria across demographic groups.

Inputs: Performance metrics (TPR, FPR, PPV) for each demographic group (Race, Gender, Income, Insurance, Geography).

Three Fairness Criteria

Criterion	Definition	Threshold (pass)
Demographic Parity	Positive prediction rate equal across groups	< 5 percentage point gap
Equal Opportunity	True positive rate equal across groups	< 10 percentage point gap
Predictive Parity	Precision (PPV) equal across groups	< 10 percentage point gap

Output: Pass/fail badge per criterion per demographic group; maximum disparity value; recommended mitigation strategy.

Tab 2 — Bias Source Analyzer

Identify likely sources of bias in the training pipeline. Checklist of 12 bias sources across data collection, labeling, model design, and deployment stages. Severity rating (High / Medium / Low) for each identified source.

Tab 3 — Mitigation Planner

Map identified bias sources to evidence-based mitigation strategies (re-sampling, re-weighting, adversarial debiasing, threshold adjustment, stratified monitoring). Estimate implementation effort (Low / Medium / High) and expected bias reduction for each strategy.

Tool 3 — AI Governance Framework Builder

Six-domain governance assessment producing a governance score and regulatory risk rating.

6 Domains, 22 Questions

Domain	Questions	Focus
Risk Classification	4	FDA SaMD classification (Class I–III); intended use; patient risk
Data Governance	4	Training data documentation; provenance; de-identification; consent
Clinical Validation	4	Prospective validation; external validation; subgroup analysis; intended population
Bias & Equity	4	Protected class analysis; disparity monitoring; corrective action plan
Monitoring & Alerting	4	Production monitoring cadence; drift detection; performance alert thresholds
Transparency & Explainability	2	Clinician-facing explanations; patient disclosure

Each question is answered Yes/No/Partial. Each has a regulatory risk impact score.

Outputs:

- Governance score (0–100) with color-coded maturity level
- Regulatory risk rating: Low / Moderate / High / Critical
- Domain-by-domain gap analysis
- Top 3 priority remediation actions with effort and risk-reduction estimates

Tool 4 — Build vs. Buy Calculator

NPV analysis comparing internal development of a clinical AI tool vs. purchasing a vendor solution.

Build Inputs: Development team cost (FTE × salary × months), infrastructure cost, integration cost, maintenance (% of build cost per year), timeline to deployment (months).

Buy Inputs: License/SaaS cost (annual), implementation cost (one-time), integration cost, vendor contract term (years).

Shared Inputs: Discount rate (%), time horizon (years), expected clinical value per year (\$).

Outputs:

- 5-year NPV for Build and Buy scenarios
- Crossover point (year at which one becomes more favorable)
- Risk-adjusted NPV (adjusting for build project failure rate)
- Recommendation: Build / Buy / Hybrid with rationale

Tool 5 — AI Medical Scribe ROI Calculator

Calculate the return on investment for deploying an AI medical documentation (ambient scribe) solution.

Inputs: Number of physicians; Average documentation time per encounter (minutes); Encounters per physician per day; Hourly physician rate (\$); AI scribe cost per physician per month (\$); Estimated documentation time reduction (%); Physician burnout reduction estimate (% FTE retention benefit).

Outputs:

- Hours saved per physician per day
- Annual documentation time saved across all physicians
- Annual time value of savings (hours × hourly rate)
- Monthly ROI and payback period
- Burnout cost offset (FTE retention × recruitment/training cost)
- Net annual benefit and ROI %

Tool 6 — Digital Health Lab

Three calculators for RPM, telehealth, and patient engagement platform ROI.

Remote Patient Monitoring (RPM) ROI Calculator

CPT code revenue model: CPT 99453 (device setup), 99454 (device supply), 99457 (first 20 min monitoring), 99458 (additional 20 min), 99091 (data interpretation).

Inputs: enrolled patient count, device type, monthly monitoring minutes, hospitalization reduction rate, payer mix.
Outputs: total annual reimbursement by CPT code, gross margin, hospitalizations avoided, net ROI.

Telehealth Utilization Modeler

Models visit volume, revenue, and cost under different CMS policy scenarios. Inputs include specialty, payer mix, visit volume, and originating site status. Outputs include revenue comparison across policy scenarios and cost-per-visit analysis.

Patient Engagement Platform Comparator

Side-by-side evaluation across activation rate, NPS score, FHIR compliance, cost PMPM, and ROI.

Part V — Interoperability & Risk

[/research-lab/interoperability](#)

Tool 1 — FHIR Interoperability Lab

Five-tab FHIR R4 testing and compliance workbench.

Tab 1 — FHIR Resource Builder

Build and validate FHIR R4 resource JSON. Select a resource type, complete the guided form, and the tool generates standards-compliant FHIR JSON.

Resources supported: Patient, Observation, Condition, MedicationRequest, Encounter, DiagnosticReport, Procedure, AllergyIntolerance, Immunization, CarePlan.

Observation LOINC Codes pre-loaded (10):

LOINC	Description
8480-6	Systolic blood pressure

LOINC	Description
8462-4	Diastolic blood pressure
8867-4	Heart rate
9279-1	Respiratory rate
8310-5	Body temperature
29463-7	Body weight
8302-2	Body height
2339-0	Glucose in Blood
4548-4	Hemoglobin A1c
2160-0	Creatinine in Serum

Output: Generated FHIR JSON with copy button; validation status (valid / warnings / errors); element-by-element validation detail.

Tab 2 — Terminology Mapper

Cross-map clinical terms across ICD-10, SNOMED CT, LOINC, and RxNorm. 15 pre-loaded conditions with cross-system code mappings.

Pre-loaded mappings (15 conditions including):

Clinical Term	ICD-10	SNOMED CT
Diabetes mellitus type 2	E11.9	44054006
Hypertension	I10	38341003
Sepsis	A41.9	91302008
Acute MI	I21.9	22298006
COPD	J44.1	13645005
Major depressive disorder	F32.9	370143000
CKD stage 3	N18.3	433144002
Atrial fibrillation	I48.91	49436004
Heart failure	I50.9	84114007
Stroke	I63.9	230690007
(+ 5 more)		

Tab 3 — CDS Hooks Tester

Simulate Clinical Decision Support (CDS) Hooks invocations and inspect the CDS card responses.

Pre-built CDS Scenarios (6):

Scenario	Hook Type	Alert Level
Drug-Drug Interaction (Warfarin + Aspirin)	order-select	Critical
Overdue HbA1c Screening	patient-view	Warning
Statin Therapy for CVD Risk	order-select	Info
Opioid High-Risk Alert	order-sign	Critical

Scenario	Hook Type	Alert Level
Sepsis Early Warning	patient-view	Critical
Preventive Care Gap	patient-view	Info

Each scenario shows: summary card text, detail, indicator level, source citation, suggestion cards (actionable order recommendations). The test shows the full CDS Hooks JSON request/response structure.

Tab 4 — Prior Authorization (FHIR PA)

Model the FHIR-based prior authorization workflow under the CMS 2024 Interoperability Rule.

PA Scenarios pre-loaded: Home health, Specialty drug, Imaging (MRI), DME, Physical therapy, Mental health inpatient.

Each scenario shows: required FHIR resources (Coverage, Claim, ClaimResponse), response time requirements (72 hours urgent / 7 days standard), and example FHIR ClaimResponse JSON.

Tab 5 — ONC Compliance Checker

Self-assessment checklist for 21st Century Cures Act compliance.

Compliance Areas: Patient access API (SMART on FHIR), Provider access API, Payer-to-payer exchange, Information blocking attestation, Electronic prior authorization, CDS Hooks support.

For each area: compliance status toggle (Compliant / Partial / Non-Compliant), penalty risk level, implementation timeline.

Output: Overall compliance score; highest-risk gaps; recommended remediation steps with ONC regulatory citations.

Tool 2 — Risk Stratification Engine

Four-tab risk scoring and population segmentation suite.

Tab 1 — HCC v28 RAF Calculator

Calculate a Medicare Advantage risk adjustment factor (RAF) score for an individual patient.

Demographic Base RAF (by age and gender) — lookup table from age 18–99, male/female. Examples:

- Age 70, Female: ~0.459
- Age 80, Male: ~0.514

20 HCC Conditions across 9 Categories with RAF Values

Category	Condition	RAF Value
Diabetes	Diabetes with Chronic Complications	0.302
Cardiovascular	Congestive Heart Failure	0.331
Cardiovascular	Coronary Artery Disease	0.150
Pulmonary	COPD	0.335
Pulmonary	Chronic Respiratory Failure	0.587
Metabolic	Morbid Obesity	0.272
Metabolic	Protein-Calorie Malnutrition	0.455
Behavioral	Major Depressive Disorder	0.309

Category	Condition	RAF Value
Behavioral	Substance Use Disorder	0.395
Renal	Chronic Kidney Disease Stage 4–5	0.289
Renal	ESRD on Dialysis	0.493
Cancer	Lung/Head/Neck Cancer	1.024
Cancer	Metastatic Cancer	2.659
Musculoskeletal	Hip Fracture/Dislocation	0.394
Musculoskeletal	Inflammatory Arthritis	0.421
Neurological	Multiple Sclerosis	0.522
Neurological	Ischemic/Hemorrhagic Stroke	0.346
Wounds	Pressure Ulcer Stage 3–4	1.233
Wounds	Diabetic Foot Ulcer	0.575
(Additional conditions)		

Total RAF = Demographic Base RAF + sum of selected condition RAFs

Risk Tier

RAF Score	Tier
< 1.0	Low
1.0–1.5	Average
1.5–2.5	High
> 2.5	Very High

Estimated MA Payment = Total RAF × \$950/month (monthly), × \$11,400 (annual)

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Tab 2 — Population Risk Segmentation

Model cost and risk for a virtual patient panel.

Inputs:

- Panel size (500–50,000 patients)
- Payer mix (Medicare / Medicaid / Commercial / Uninsured %)
- Risk tier distribution (Very Low / Low / Moderate / High / Very High % — must sum to 100%)
- Condition prevalence sliders (Diabetes, CHF, COPD, CKD, Depression, Substance Use)
- SDOH risk factors (Housing instability, Food insecurity, Transportation barriers)

Annual Cost Benchmarks by Tier (AHRQ MEPS / CMS claims):

Tier	Annual Cost per Patient
Very Low	\$1,800
Low	\$3,200
Moderate	\$8,500
High	\$22,000
Very High	\$68,000

Outputs:

- Total annual panel cost and cost per patient
- High-risk patient count and % of panel
- Potential savings if 10% of High/Very High tier moves to Moderate (\$cost avoided)
- SDOH Complexity Index (% of panel at elevated social risk)
- Segmentation bar chart

Tab 3 — Custom Risk Model Builder

Build a custom risk scoring model using selected clinical, demographic, and SDOH variables.

Variables: Age, Chronic condition count, Prior hospitalization, ER utilization, Medication adherence score, SDOH score, Disability status, Dual eligibility.

Inputs: Weight each variable (0–10); define risk tier thresholds.

Output: Risk score distribution for a sample panel; sensitivity analysis of weight changes; comparison to HCC RAF alignment.

Tab 4 — Comorbidity Interaction Analyzer

Model the compounding cost and risk effect of multi-condition comorbidity combinations.

Pre-loaded comorbidity pairs and triads: CHF+CKD, Diabetes+CKD, Diabetes+Depression, CHF+Depression+CKD, (+ others).

Output: Additive vs. synergistic cost multiplier; RAF interaction value; recommended care management intensity.

Part VI — Knowledge & Workspace

/research-lab/knowledge-workspace

Tool 1 — Evidence Library

Searchable, filterable reference library containing 25 CEA studies, 20 CMMI innovation models, and 15 HTR policy briefs.

Section A — CEA Evidence Library (25 studies)

Peer-reviewed cost-effectiveness analyses from NEJM, JAMA, Lancet, and other major journals. Each entry includes: title, journal, year, condition, intervention, ICER value, verdict, and evidence level.

Sort options: ICER (ascending/descending), Year, Journal, Title

Filter by: Condition area, Verdict, Evidence level (High / Moderate / Low)

Verdict classifications:

Verdict	ICER Range
Dominant	ICER < \$0 (saves money AND improves health)
Highly Effective	< \$30,000/QALY
Cost-Effective	\$30,001–\$100,000/QALY
Borderline	\$100,001–\$150,000/QALY
Not Cost-Effective	> \$200,000/QALY

Selected studies (highlights):

Study	ICER	Verdict
Metformin for T2D Prevention (NEJM 2002)	\$11,000	Highly Effective
Buprenorphine for OUD (JAMA 2019)	\$13,500	Highly Effective
Childhood MMR Vaccine (MMWR 2014)	−\$500	Dominant
Hospital at Home (Ann Int Med 2022)	−\$14,000	Dominant
Palliative Care Consultation (NEJM 2010)	−\$8,000	Dominant
Lung Cancer Screening LDCT (NEJM 2011)	\$52,000	Cost-Effective
GLP-1 Agonists for Obesity (NEJM 2021)	\$175,000	Borderline
PCSK9 Inhibitors (NEJM 2017)	\$325,000	Not Cost-Effective
CAR-T Therapy Kymriah (NEJM 2018)	\$610,000	Not Cost-Effective
Gene Therapy Zolgensma (NEJM 2019)	\$900,000	Not Cost-Effective

Section B — CMMI Innovation Model Tracker (20 models)

Track all major Center for Medicare and Medicaid Innovation models: status, participants, savings evidence, and key lessons.

Model Status: Active, Ended, Cancelled

Model Types: ACO, Episode, Primary Care, Specialty, State/Rural, State/Equity, State/All-Payer, State/Pediatric

20 Models Tracked (selected):

Model	Status	Participants	Key Finding
MSSP	Active	485 ACOs	\$1.8B net savings through 2022; Enhanced track outperforms
ACO REACH	Active	150	Successor to Direct Contracting; mandatory health equity plans
BPCI-Advanced	Active	1,000	\$600M gross savings; orthopedic/cardiac working well
Vermont All-Payer ACO	Active	1 (state)	Only all-payer state model; mixed early results
MSSP Next Gen ACO	Ended	53	Too complex; replaced by ACO REACH
CPC+	Ended	3,000	No net savings despite quality improvement; 5-year timeline too short
Independence at Home	Ended	14	\$2,700 savings/beneficiary; scalability limited by workforce
GUIDE (Dementia Care)	Active	400	Comprehensive dementia model with caregiver support
Making Care Primary (MCP)	Active	200	10-year primary care transformation pathway
Transforming Maternity Care	Active	150	First bundled maternity model with doula integration
Radiation Oncology Model	Cancelled	0	Cancelled after stakeholder opposition

Each model entry expands to show full lessons learned (100–200 word analysis per model).

Section C — HTR Policy Briefs (15 briefs)

In-depth HTR-authored policy analyses on major healthcare transformation topics. Each brief includes: title, area, key statistic, 150-word abstract, and 500-word full body.

Topic Areas: Payment Reform, Access, Health IT, State Policy, Workforce, Equity

15 Briefs include:

Title	Key Statistic
All-Payer Claims Databases: State Policy Landscape 2024	\$0 federal support for APCD infrastructure
Medicaid Expansion: 10-Year Impact Assessment	4.1M adults still in coverage gap
Site-Neutral Payment: Evidence For and Against	Hospital outpatient pays 2.2× more than physician office
FHIR Interoperability: Status 2025	91% of hospitals have certified FHIR R4 APIs
Medicare Advantage: Overpayment and Risk Coding	MedPAC estimates \$88B cumulative overpayment
Vermont Global Budget: Lessons Learned	GMCB regulated \$4.2B in hospital spending in FY2024
AI Regulation in Healthcare: Federal Policy Gap Analysis	FDA cleared 882 AI/ML devices through 2023
Prior Authorization Reform: Legislative Landscape	1 in 4 physicians report a patient harm from PA delays
No Surprises Act: Implementation Assessment	490,000+ surprise billing disputes in first 18 months
(+ 6 additional briefs)	

Tool 2 — Workforce Modeler

Four-tab healthcare workforce supply, demand, and retention modeling tool.

Tab 1 — Physician Supply & Demand

10-year national or regional physician supply and demand projection.

12 Specialties pre-loaded:

Specialty	Current FTE	Annual Grads	Retirement Rate	Demand/Provider
Family Medicine	109,000	4,800	3.5%	3,029 patients
Internal Medicine	135,000	8,000	3.5%	2,444 patients
General Surgery	26,000	1,200	3.8%	12,692 patients
Orthopedics	20,700	800	3.2%	15,942 patients
Cardiology	22,000	2,000	4.0%	15,000 patients
Psychiatry	33,000	1,900	4.5%	10,000 patients
Emergency Medicine	41,000	2,200	3.0%	8,049 patients
OB/GYN	19,000	1,300	4.2%	8,684 patients
Pediatrics	33,000	2,800	3.5%	2,242 patients
Oncology	13,300	700	3.8%	24,812 patients

Specialty	Current FTE	Annual Grads	Retirement Rate	Demand/Provider
Neurology	16,700	900	3.5%	19,760 patients
Radiology	34,000	1,100	3.3%	9,706 patients

Adjustable Inputs: Current FTE, Annual graduates, Retirement rate (%), IMG contribution, Population served, Demand per provider.

Geographic Scope: National (1.0×), Northeast (0.18×), Southeast (0.22×), Midwest (0.21×), West (0.24×), Rural (0.14×), Urban (0.56×).

Outputs: 10-year supply projection (FTE per year accounting for retirements + graduates + IMG); demand projection; supply/demand gap by year; shortage or surplus classification; graph of supply vs. demand curves.

Tab 2 — Nurse Staffing Ratios

Model the cost and FTE impact of implementing mandatory nurse-to-patient staffing ratios.

7 Unit Types pre-loaded:

Unit	Current Ratio	Mandated Ratio	Avg Salary	Agency %
ICU	1:2	1:2	\$105,000	15%
Med/Surg	1:6	1:5	\$82,000	20%
Emergency Dept	1:4	1:4	\$95,000	25%
Labor & Delivery	1:2	1:2	\$98,000	12%
Pediatrics	1:4	1:4	\$88,000	18%
Telemetry	1:5	1:4	\$86,000	22%
Behavioral Health	1:6	1:5	\$78,000	30%

Inputs: Current ratio, mandated ratio, census (occupied beds), current FTE, average salary, agency nurse proportion (%).

Outputs:

- Additional FTEs required to meet mandated ratio
- Annual incremental labor cost
- Agency cost if filled with travel nurses vs. permanent hire
- Annualized cost per unit type
- Total hospital cost of mandate compliance

Tab 3 — Turnover ROI Calculator

Calculate the total cost of staff turnover and the ROI of retention investments.

Role Defaults pre-loaded:

Role	Salary	Turnover Rate	Recruit Cost	Onboard Cost	Vacancy Days	Agency Premium
RN	\$85,000	22%	\$5,000	\$10,000	75	120%
Physician	\$280,000	8%	\$50,000	\$80,000	120	150%
APP (NP/PA)	\$130,000	14%	\$20,000	\$25,000	90	130%
Medical Assistant	\$42,000	35%	\$2,000	\$3,000	30	80%

Role	Salary	Turnover Rate	Recruit Cost	Onboard Cost	Vacancy Days	Agency Premium
(+ additional roles)						
Inputs: Role selection, number of FTEs in role, current turnover rate, recruit cost, onboarding cost, vacancy days, agency fill premium.						
Outputs: - Total annual turnover cost - Cost per departure (recruit + onboard + vacancy + agency) - Breakeven investment in retention programs - ROI of 1% turnover reduction						

Tab 4 — Rural Workforce Shortage Analysis

Model rural healthcare workforce gaps and the financial case for rural-specific incentive programs.

Inputs: Rural population size; current provider-to-population ratio; target ratio; loan forgiveness program cost; retention bonus cost; rural training pipeline investment.

Outputs: Current shortage (FTEs needed); annual population-level impact of shortage; cost of filling gap with each incentive type; 10-year ROI of rural pipeline investment.

Tool 3 — Innovation Leaderboard

Ranked leaderboards showing healthcare transformation performance across states, health systems, and payers.

Three Leaderboards:

State Transformation Rankings — 50 states scored and ranked on: AHEAD/APM adoption rate, quality measure performance, uninsured rate trajectory, Medicaid expansion status, SDOH investment index, hospital global budget adoption.

Health System Performance — Rankings across major health systems on: value-based care revenue %, readmission rate, quality composite, patient experience score, cost efficiency index, equity performance.

Payer Innovation Index — Rankings across major payers/insurers on: APM adoption rate, Star Rating, member HEDIS composite, prior authorization denial rate, network adequacy score.

Filter options: Region, System type (academic / community / rural / safety-net), Payer type (Commercial / Medicaid / Medicare).

Tool 4 — Research Workspace

Save, organize, and export your Research Lab work across sessions.

Features:

Scenario Storage — Save named scenarios from any lab tool. Each saved scenario stores all input parameters so you can reload and compare configurations.

Compare View — Side-by-side comparison of up to 4 saved scenarios from the same tool. Differences between scenarios are highlighted.

Report Templates — Three pre-built report formats:

- Strategic Summary (executive-level, 1 page)
- Technical Analysis (full inputs + outputs + methodology notes)
- Policy Brief (framing, evidence synthesis, recommendation structure)

Citation Generator — Auto-generates citations for the data sources and evidence underlying each tool's pre-loaded parameters (AHA, AHRQ, CMS, NEJM, JAMA, etc.).

Export Options — Download workspace as PDF report or CSV data file.

Note: The Research Workspace uses browser local storage. Data persists across page reloads on the same browser but does not sync across devices or browsers. Export your work before clearing browser storage.

Using the Research Lab Effectively

These tools are for scenario modeling, not prediction. All outputs are based on published evidence-based parameters and user-supplied inputs. They are designed to inform strategic planning, build the business case for program investment, and support evidence-based conversations — not to produce final financial forecasts for board-level budget commitments without additional validation.

Vermont-specific calibration. The Vermont All-Payer ACO Model data, Global Budget Simulator, and 1115 Waiver Modeler are explicitly calibrated to Vermont policy contexts. Vermont-specific prevalence rates, cost benchmarks, and regulatory parameters should be used when available to replace national defaults.

Saving your work. Tools do not persist state between browser sessions unless you use the Research Workspace save function. Screenshot or export results before closing.

Data sources. Pre-loaded parameter values are sourced from: AHA Annual Survey, AHRQ MEPS, CMS National Health Expenditure Accounts, NCQA HEDIS benchmarks, MedPAC reports, Kaufman Hall National Hospital Flash Report, AAMC workforce data, and peer-reviewed literature. Citations are available in the Research Workspace citation generator.

Model limitations. The HEDIS simulator, APM Calculator, and Hospital Financial Scorecard use simplified calculation models calibrated to real-world data but are not certified actuarial tools. For regulatory submissions, contract negotiations, or GAAP financial reporting, engage a licensed actuary.

Advisory escalation. If your modeling reveals a scenario requiring deep customization — multi-state APM design, Act 167 simulation for specific facilities, or population health program implementation — the Advisory Services panel on each lab page links directly to the HTR Advisory team.